

A PRACTITIONER BRIEFING

The Earned Value Management Playbook

From first principles to proactive, AI-enabled program control

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Grounded in EIA-748, NDIA IPMD, and NASA EVM guidance · Original work, frameworks cited to source

Executive Summary

Most organizations do not fail their major programs for lack of effort or budget. They fail because they discover the overrun too late to act on it. Earned Value Management (EVM) exists to close that gap: it is the discipline of measuring **work actually accomplished** — not money spent or time elapsed — so that cost and schedule problems surface months earlier, while they are still cheap to fix.

This playbook is a working reference for the practitioner: the control account manager, the project controls analyst, the planner/scheduler, and the program manager who owns an EVM requirement and wants to **use** it rather than merely comply with it. It moves deliberately from first principles to advanced practice, and it closes with the shift now reshaping the field — the use of **agentic AI** to automate reporting and to hunt emerging risk proactively across large, long-duration portfolios.

Three convictions run through everything that follows:

1. **Spending is not progress.** The single most expensive error in program management is treating money burned or time elapsed as evidence of accomplishment. EVM's entire value is refusing to make that mistake.
2. **Variances are signals, not verdicts.** A variance is an early-warning indicator that invites investigation — not an occasion for blame, and never a number to be engineered away.
3. **The baseline is sacred.** You may replan the future; you may never rewrite the past. The integrity of the measurement baseline is the practitioner's core professional responsibility.

Bottom line. EVM converts a program from a rear-view mirror, reviewed monthly, into a forward-looking radar. The organizations that win are those that act on the signal early — and, increasingly, those that let AI watch the radar continuously.

The Core Thesis: Why Traditional Tracking Fails

The two questions that mislead

Conventional program tracking rests on two comparisons: schedule (today's date versus the plan) and budget (money spent versus money budgeted). Both are necessary and neither is sufficient. Consider a program that, at the midpoint of its schedule, has consumed exactly half its budget. Conventional tracking reports "on plan." Yet if only one-quarter of the scope is complete, the program is on a trajectory toward a substantial overrun that neither metric can detect.

The failure is structural, not a matter of diligence. Spend rate and calendar position say nothing about **how much of the work is actually finished**. A program can consume

resources at exactly the planned rate while producing far less than planned — and conventional tracking will applaud it right up to the crisis.

The corrective: a third dimension

EVM introduces the missing measurement: the budgeted value of work **actually performed**. With that single addition, the practitioner can compare three quantities that are finally commensurable — what was planned, what was accomplished, and what was spent — and thereby integrate the three axes every program manages: scope, schedule, and cost.

Management question	Traditional answer	EVM answer
On schedule?	Date vs. plan	Work done vs. work planned
On budget?	Spent vs. planned	Spent vs. work accomplished
Actually progressing?	Cannot determine	The central measurement

The Three Primary Measurements

Every EVM metric derives from three time-phased quantities. The practitioner should internalize both their modern names and the classical EIA-748 terminology, which remains in wide use across government and industry contracts.

Planned Value (PV / BCWS)

Planned Value is the budgeted cost of the work scheduled to be complete as of the status date — the measurement baseline against which everything else is compared. Its classical name is the **Budgeted Cost of Work Scheduled (BCWS)**; the two are identical.

Earned Value (EV / BCWP)

Earned Value is the budgeted cost of the work **actually completed**, valued at the budget originally assigned to that work. Its classical name is the **Budgeted Cost of Work Performed (BCWP)**.

The critical subtlety: EV is priced at **budgeted** cost, never at actual cost. Completing an activity budgeted at \$20,000 earns exactly \$20,000 of value, regardless of whether it cost \$18,000 or \$25,000 to accomplish. This deliberate design keeps EV a pure measure of accomplishment, uncontaminated by spending performance.

Actual Cost (AC / ACWP)

Actual Cost is the cost incurred to accomplish the work performed, drawn from the accounting system and aligned to the same status cutoff. Its classical name is the **Actual Cost of Work Performed (ACWP)**.

Bottom line. At any status date, the vertical distances among PV, EV, and AC constitute the program's entire performance narrative. Every metric in the next section quantifies one of those gaps.

Deriving Insight: Variances and Indices

Four derived metrics turn the three measurements into decisions — two variances (absolute, in currency) and two indices (dimensionless ratios centered on 1.0).

Bottom line. Interpretation rule: for variances, negative is unfavorable; for indices, below 1.0 is unfavorable.

Cost Variance (CV) and Cost Performance Index (CPI)

Cost Variance measures cost efficiency in absolute terms; the **Cost Performance Index** expresses the same as a ratio.

$$\begin{aligned} \text{CV} &= \text{EV} - \text{AC} && (\text{negative} \Rightarrow \text{overrun}) \\ \text{CPI} &= \text{EV} \div \text{AC} && (\text{below } 1.0 \Rightarrow \text{overrun}) \end{aligned}$$

A CPI of 0.80 means the program realizes 80 cents of budgeted work for every dollar spent. Read forward, that efficiency implies a material overrun at completion unless the trend reverses.

Schedule Variance (SV) and Schedule Performance Index (SPI)

Schedule Variance and the **Schedule Performance Index** measure schedule efficiency in the currency of work.

$$\begin{aligned} \text{SV} &= \text{EV} - \text{PV} && (\text{negative} \Rightarrow \text{behind schedule}) \\ \text{SPI} &= \text{EV} \div \text{PV} && (\text{below } 1.0 \Rightarrow \text{behind schedule}) \end{aligned}$$

A caution that separates the fluent practitioner from the novice: SV and SPI are denominated in **dollars of work**, not units of time. An SPI of 0.90 does not mean “ten percent late in days.” It means ten percent of the planned **value** is not yet earned. Translating that into calendar impact requires the Integrated Master Schedule and its critical path. EVM establishes **that** a program is behind; the schedule quantifies **by how much**.

Bottom line. The ten-second read: given any two of CV, SV, CPI, SPI, a practitioner should be able to state cost and schedule health in a single sentence and name which one demands attention first.

Applied Analysis: A Worked Program

To ground the metrics, we use the ACME construction program from the standard training curriculum. The program comprises four scope elements at its first status review. All figures are in dollars.

Scope element	PV (planned)	EV (earned)	AC (actual)
Foundation	15,394	15,394	15,850
Patio	8,166	8,166	7,200
Exterior Walls	8,748	6,608	6,250
Stairway	5,961	2,981	3,100
Program total	38,269	33,149	32,400

Applying the four metrics at the program level:

- **CV** = $33,149 - 32,400 = +749$ — a modest cost underrun.
- **SV** = $33,149 - 38,269 = -5,120$ — a meaningful schedule shortfall.
- **CPI** = $33,149 \div 32,400 = 1.02$ — two percent under budget.
- **SPI** = $33,149 \div 38,269 = 0.87$ — approximately thirteen percent behind schedule.

The headline: **the program is two percent under budget but thirteen percent behind schedule**. The professional discipline, however, is to decompose the total and locate the drivers.

Scope element	CV	SV	CPI	SPI
Foundation	-456	0	0.97	1.00
Patio	+966	0	1.13	1.00
Exterior Walls	+358	-2,140	1.06	0.76
Stairway	-119	-2,980	0.96	0.50
Program total	+749	-5,120	1.02	0.87

The decomposition tells a precise story. The cost underrun is driven almost entirely by the Patio (CPI 1.13). The schedule shortfall is concentrated in the Stairway (SPI 0.50 — only half its planned work is complete) and the Exterior Walls (SPI 0.76). The actionable conclusion is not “the program is a little behind”; it is “redirect attention and, if warranted, resources to the Stairway and Exterior Walls, whose delays account for essentially the entire variance.”

Materiality thresholds

Practitioners do not write a formal explanation for every deviation. Most organizations set variance thresholds between plus-or-minus seven and ten percent; a CPI or SPI outside the 0.90–1.10 band typically triggers a written variance analysis. Favorable variances

warrant explanation as rigorously as unfavorable ones — an unexpected underrun may signal an estimating error or a deferred scope problem, not a genuine efficiency.

Constructing the Baseline

Performance cannot be measured against a plan that does not exist. The baseline is built in a fixed sequence, each layer depending on the one beneath it.

Work Breakdown Structure

The **Work Breakdown Structure (WBS)** is a deliverable-oriented decomposition of the total scope into progressively smaller work elements. Two disciplines govern it. First, the WBS describes **work**, not organization — it is not an org chart; the **Organizational Breakdown Structure (OBS)** captures who performs the work, and the intersection of the two is expressed in the **Responsibility Assignment Matrix (RAM)**. Second, the **100-percent rule**: the children of any element must sum to exactly the parent — no omitted scope, no invented scope. Each element carries a one-line definition in the **WBS dictionary**.

A practical heuristic: decompose slightly deeper than current reporting requires. Detailed elements roll up to a summary trivially; a coarse baseline cannot be split into detail once work is underway.

Schedule

The scheduled activities, sequenced by logical dependencies, form a network. Durations are estimated — the **Critical Path Method (CPM)** uses a single most-likely estimate; the **Program Evaluation and Review Technique (PERT)** blends optimistic, likely, and pessimistic estimates. A forward and backward pass then yields each activity's early and late dates. Two concepts are indispensable: the **critical path** (the longest dependent chain, which sets the completion date) and **float** (the delay an activity can absorb without moving the finish). All of this resides in the **Integrated Master Schedule (IMS)**, the single authoritative source of schedule data.

Budget and control accounts

Budget is time-phased onto the scheduled work. The **control account** — where a WBS element meets a single responsible organization — is the fundamental unit of EVM: the point at which planned value is set, earned value is claimed, and actual cost is collected. Each is owned by one **Control Account Manager (CAM)**.

The sum of all time-phased control-account budgets is the **Performance Measurement Baseline (PMB)** — the PV curve against which performance is measured. Two related terms recur constantly: the **Budget at Completion (BAC)**, the total authorized budget for the work (which must equal total PV), and **Management Reserve (MR)**, budget held by the program for identified risks, inside the total budget but outside the PMB.

Earned value techniques

The method by which value is claimed is fixed in advance per work package:

- **Fixed formula** (0/100, 50/50, 25/75) — for short-duration packages; a defined fraction is earned at start and the remainder at completion.
- **Milestone weighting** — value earned only upon achieving pre-defined milestones; suited to longer packages.
- **Percent complete** — flexible but the most susceptible to subjectivity; requires discipline.
- **Units complete** — highly objective, for repetitive production work.
- **Level of effort (LOE)** — earns value with the passage of time; appropriate only for support work with no discrete product, and used sparingly because it can mask problems.

Bottom line. The governing principle across all techniques: value is claimed for objective, verifiable accomplishment — never for effort, and never for a subjective “ninety percent complete” that persists unchanged for weeks.

Forecasting Program Outcomes

Performance to date is the most reliable available predictor of performance to come, and EVM makes that prediction quantitative. The **Estimate to Complete (ETC)** is the forecast cost of the remaining work; the **Estimate at Completion (EAC)** is the forecast total cost — actuals to date plus ETC. (The **Latest Revised Estimate**, LRE, is a synonym for the current EAC.)

The most common independent forecast assumes cost efficiency to date persists:

$$\text{EAC} = \text{BAC} \div \text{CPI}$$

For a program with a \$10M BAC running at a CPI of 0.80, the statistical EAC is \$12.5M — a \$2.5M projected overrun visible in the data long before it would appear in an accounting close. More sophisticated forms weight both cost and schedule efficiency (for programs where schedule slippage itself drives cost). The discipline is not to memorize every variant but to **state explicitly which assumption a given EAC embodies** when briefing leadership.

A final forward-looking check is the **To-Complete Performance Index (TCPI)** — the efficiency required across all remaining work to still achieve the target. A TCPI of 1.30 set against a demonstrated CPI of 0.80 is the arithmetic declaring the target implausible: it asks a team that has performed at 80 percent to suddenly perform at 130 percent.

Integrating Risk and Opportunity

The distinction between a competent reporter and a genuine practitioner is this: EVM is a risk-management instrument, not a scorecard. The professional guidance is explicit — an unfavorable variance frequently quantifies a **realized risk**, while a favorable variance often represents a captured **opportunity** (the ACME Patio underrun, for instance, frees budget that can be returned to Management Reserve).

Risk and EVM are meant to operate as one system. In planning, risk assessment sets the level of Management Reserve and its allocation across control accounts. In execution, EVM trends reveal whether risk-handling plans are working. The decisive skill is reading **trends**, not single periods: an isolated adverse month is noise, but a control account whose CPI drifts from 0.98 to 0.94 to 0.91 across three periods is a realized risk announcing itself in slow motion. On multi-year programs, detecting that drift a single quarter earlier is frequently the difference between a routine correction and a crisis.

Bottom line. The structural limitation is scale. A large program carries dozens of control accounts and hundreds of work packages. No individual can inspect every one, every period, for a subtle three-period drift. This blind spot is precisely what the AI section addresses.

Change Management and Baseline Integrity

Change is inevitable; corrupting the performance record is not. Three categories of change must be distinguished:

- **Replanning** — adjusting **future** work within the current budget. Routine and legitimate (a design change, a resource re-profile, issuing MR to fund a newly identified risk).
- **Rebaselining / reprogramming** — a formal reset when the program can no longer be executed within its targets. A significant action requiring customer authorization and, typically, a fresh Integrated Baseline Review.
- **Prohibited** — altering **historical** performance to erase a variance or achieve a target index. The moment replanning becomes a tool to eliminate red, the system forfeits its objectivity and the organization falls out of EIA-748 compliance.

Bottom line. The inviolable rule: replan the future; never rewrite the past. Maintain a complete record of the original baseline and every subsequent change.

Agentic AI for the EVM Practitioner

The monthly cadence exposes a durable inefficiency. Assembling data, computing metrics, and drafting narratives consume the analyst's time; judgment, corrective action, and stakeholder engagement — the activities that create value — receive whatever remains.

And as the risk section established, no human can proactively scan every control account for slow-moving trends. Both constraints are now addressable.

What “agentic” means here

A conventional assistant answers questions on request. An **agent** is given an objective and access to tools, and executes an end-to-end task autonomously — retrieving data, performing analysis, validating results, and drafting outputs — escalating to a human only where a decision is required. Applied to EVM, such an agent is granted read access to the cost tool, the IMS, the accounting feed, and the risk register, under a standing objective: keep the program ahead of every emerging cost and schedule problem.

Use case one — automated status reporting

Rather than an analyst spending days assembling the monthly package, the agent executes the full cycle: it retrieves PV, EV, and AC for every control account at the cutoff; computes CV, SV, CPI, and SPI and re-forecasts EAC and ETC across the WBS; and **drafts the variance analyses** — what moved, why (by correlating schedule slips and risk-register entries), and the at-completion impact — in the organization’s required format, whether IPMDAR-aligned or an internal template. A single engine produces two cadences: a concise weekly pulse and the full monthly management-review package. The practitioner’s role shifts from **producing** the report to **reviewing, correcting the judgment calls, and signing** it. Days compress to minutes, and output becomes consistent period over period.

Use case two — proactive risk and opportunity deep-dives

This is the higher-value application, and it directly closes the scale gap. Operating continuously between reporting cycles, the agent:

- **scans every control account for drift** — the incremental CPI erosion that escapes human notice because each individual period appears acceptable;
- **detects leading indicators** — critical-path float eroding, ETCs creeping, actuals outpacing earned value;
- **conducts the causal deep-dive automatically** — correlating EVM data with the schedule and risk register, forming a hypothesis about root cause, quantifying the projected EAC impact if the trend persists, and drafting a recommended corrective action or MR draw;
- **surfaces opportunities as well as threats** — flagging favorably trending accounts as candidates to replan and return budget to reserve;
- **escalates with a complete brief** — delivering, weeks ahead of the reporting cycle, a finding such as: “Control Account 3.2.1 is trending toward a \$180K overrun by Q3; probable cause is the vendor slip logged last week; recommend acting now.”

On a multi-year program, advancing a warning by a single quarter routinely determines whether the response is an inexpensive adjustment or a formal reprogramming.

Governance and guardrails

Agentic AI amplifies the practitioner; it does not assume the practitioner's accountability. Three constraints are non-negotiable. First, the agent may replan the future but must never alter historical performance — the baseline-integrity rule applies without exception. Second, authoritative outputs require a human signature: the agent drafts, but the CAM and PM own the variance analysis, the EAC, and any baseline change. Third, every output must be fully traceable to source data with an auditable trail — the same standard EVM already demands of people.

Bottom line. Deployed within these guardrails, EVM ceases to be a monthly rear-view mirror and becomes a continuous forward-looking radar. The transition from reporting what has occurred to preventing what is about to is the defining evolution of the discipline — and it is available today.

Practitioner's Reference Checklist

- I can articulate PV, EV, and AC and the distinction between spending and accomplishment.
- Given any two of CV, SV, CPI, SPI, I can state program health in one sentence and name the priority.
- I know the program BAC and can compute a first-order EAC as $BAC \div CPI$.
- I can identify my control accounts and their CAMs.
- My PMB is one I would defend at an Integrated Baseline Review — realistic, traceable, risk-informed.
- I treat variances as early-warning signals to investigate, never as blame.
- I analyze multi-period trends, not single-period noise, especially on long-duration accounts.
- I operate EVM and the risk register as one integrated system.
- I replan the future and never rewrite the past.
- I know which elements of the cycle an AI agent should automate — and which judgments it must never own.

Glossary of Terms and Acronyms

Acronym	Term	Meaning
EVM	Earned Value Management	Managing by measuring accomplished work, not spend
EVMS	EVM System	The tools and processes that operate EVM
PV / BCWS	Planned Value	Budgeted cost of work scheduled by the status date

EV / BCWP	Earned Value	Budgeted cost of work actually performed
AC / ACWP	Actual Cost	Cost incurred for the work performed
CV	Cost Variance	$EV - AC$; negative denotes overrun
SV	Schedule Variance	$EV - PV$; negative denotes behind schedule
CPI	Cost Performance Index	$EV \div AC$; below 1.0 denotes overrun
SPI	Schedule Performance Index	$EV \div PV$; below 1.0 denotes behind schedule
BAC	Budget at Completion	Total authorized budget for the work
EAC	Estimate at Completion	Forecast total cost = actuals + ETC
ETC	Estimate to Complete	Forecast cost of the remaining work
TCPI	To-Complete Performance Index	Efficiency required on remaining work to hit target
LRE	Latest Revised Estimate	Synonym for the current EAC
WBS	Work Breakdown Structure	Deliverable-oriented decomposition of scope
OBS	Organizational Breakdown Structure	Decomposition of performing organizations
RAM	Responsibility Assignment Matrix	Maps WBS scope to responsible organizations
PMB	Performance Measurement Baseline	Time-phased budget plan; the PV curve
MR	Management Reserve	Risk budget inside total, outside the PMB
CAM	Control Account Manager	Single owner of a control account
IMS	Integrated Master Schedule	Authoritative source of all schedule data
IBR	Integrated Baseline Review	Formal review that the baseline is realistic
LOE	Level of Effort	Support work that earns value with time
CPM	Critical Path Method	Scheduling using one estimate per activity
PERT	Program Evaluation and Review Technique	Scheduling blending three estimates
EIA-748	EVMS Standard	The guideline standard a compliant EVMS meets

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Prepared by Nash Vedula, Founder of Roverax.ai — building agentic AI that turns project controls from a rear-view mirror into a forward-looking radar. This is original work; frameworks and worked examples are grounded in EIA-748, the NDIA IPMD guidance, and the NASA EVM Implementation Handbook. Distributed freely with attribution.